



TIPS & TRICKS

Auto mounting a partition in Linux

 The file that contains the data regarding the devices to be mounted at start-up is in `/etc/fstab`. To automatically mount a partition, follow the steps given below.

First, create the directory in which your partition will be mounted. Create one directory per partition. I created the directory in `/media`. This directory is known as the 'mount point' for the partition.

To create the mount point, open up the terminal and type the following command:

```
sudo mkdir location_of_dir/name_of_dir
```

...or you can use Nautilus, the file manager, to create a folder.

If the directory is created in a location in which you need root privileges, use `sudo`. After creating the mount point, modify `/etc/fstab` as per your requirements. It is always advisable to create a backup of the `/etc/fstab` file before making any changes, because any error in that file can prevent your OS from booting.

Now, make changes in `fstab` to auto mount the HDD partition:

```
sudo gedit /etc/fstab
```

Open the `/etc/fstab` with a text editor of your choice with root privileges.

In this file, add the details in the same order as done for the existing partitions.

The order should be as follows: the device name, the default mount point, the filesystem type, mount options, dump, and the `fsck` option.

The device name is the name of the HDD partition (such as `/dev/sda5`): the mount point is the full path of the directory where the partition is to be mounted. The file system type is the type of file system like `ext4`, `fat`, `ntfs`, etc. Mount options are normally given as defaults, while `dump` and `fsck` options are given as 0.

I had a partition `/dev/sda5` and I created the directory /

`media/mydisk`. My partition was of type `ext4`, so to my `/etc/fstab`, I added the following command:

```
/dev/sda5 /media/mydisk ext4 defaults 0 0
```

Save the file and in the command prompt, type the following:

```
sudo mount -a
```

Now, the partition will be automatically mounted on every reboot.

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Creating a virtual file system

 Here is a simple tip that allows you to create a virtual file system and mount it with a loopback device.

STEP 1: First create a file of 10 MB using the following command:

```
$ dd if=/dev/zero of=/tmp/disk-image count=20480
```

By default, `dd` uses a block of 512 so the size will be 20480*512

STEP 2: Now create the file system as `ext2` or `ext3`

Here, in the following example, let's use `ext3` as a file system:

```
$ mkfs -t ext3 -q /tmp/disk-image
```

You can even use `Reiser` as a file system type, but you'll need to create a bigger disk image. Something like what's shown below:

```
$ dd if=/dev/zero of=/tmp/disk-image count=50480
```

```
$ mkfs -t reiserfs -q /tmp/disk-image
```

STEP 3: As the final step, create a mount point and